

Seok (Daniel) Kim

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Education

Yonsei University

Seoul, Korea

B.S. in Computer Science & Applied Statistics

Expected Graduation: 2027

- **Key Coursework:** Reinforcement Learning, AI Systems, Machine Learning, Mathematical Statistics, **Convex Optimisation**, Stochastic Processes, Differential Privacy, Distributed Learning, Mathematics for Deep Learning, Sampling Theory, High-dimensional Data Analysis.

Technical Skills

- **Machine Learning:** Bayesian Inference (SGLD, MCMC), Variational Inference, Uncertainty Quantification.
- **Privacy-Preserving AI:** Differential Privacy (DP-SGLD), Rényi Differential Privacy (RDP), Machine Unlearning (SISA, Certified Deletion), Moments Accountant.
- **Reinforcement Learning:** Offline RL (IQL, CQL), Ray Rllib, Gymnasium, MuJoCo, Thompson Sampling, Bayesian Regret Analysis.
- **Languages:** Python (PyTorch), R, C/C++, Java, SQL, L^AT_EX (Advanced).
- **Frameworks & Tools:** PyTorch, Ray RLLib, Gymnasium, Opacus, TensorBoard, GPyTorch, Git.
- **Certificates:** SQL Developer (SQLD), Advanced Data Analytics Semi-Professional (ADsP).

Research & Capstone Experience

AI-ISL (Information Systems Laboratory), Yonsei University

Dec 2025 – Mar 2026

Undergraduate Research Assistant (Advised by Prof. Albert No)

- Contributing to the “**Preserve then Quantise (PTQ)**” project; investigating information-theoretic bounds to preserve the distributional properties of model weights during post-training quantisation.

Optimising Contextual Sentence Prediction for AI-Assisted Authoring (Episod)

Jul 2025 – Dec 2025

Lead Evaluator

- Developed LLM-as-judge rubrics for Gemini-2.5 to quantify coherence and creativity in long-form writing; conducted scoring calibration using 110+ structured test cases.

Selected Projects

Trustworthy Offline RL: Trajectory DP, IQL, and Machine Unlearning

Jun 2026

- Developed a domain-agnostic offline RL framework using **Implicit Q-Learning (IQL)** to maintain numerical stability and prevent value explosion under heavy **Rényi Differential Privacy (RDP)** noise.
- Engineered a **Trajectory-level DP** mechanism; mathematically proved that transition-level clipping understates sensitivity due to non-zero temporal covariance ($\text{Cov}(g_t, g_{t+k})$) in longitudinal records.
- Implemented certified data deletion via **LiSSA-based Influence Function Unlearning** and **SISA Sharding**, enabling approximate inverse-Hessian corrections without full model retraining.
- Validated cross-domain utility on **MIMIC-III (Healthcare)** and **FinRL (Finance)** proxies, achieving a tight privacy budget ($\epsilon \approx 2.74$) with minimal utility gap ($\Delta J \approx 0.98$ for finance).
- **Tech Stack:** PyTorch, IQL, Trajectory DP-SGD, LiSSA (Second-order Optimisation), MIMIC-III, FinRL.

Bayesian RL Meets MCMC: Sample Efficiency & Privacy-Preserving Policy Optimisation

May 2026

- Developed a principled Bayesian Reinforcement Learning (RL) framework using **Variational Actor-Critic (VAC)** and **SGLD** to mitigate Maximum Likelihood Estimation (MLE) collapse in data-scarce regimes.
- Engineered a **Differential Privacy (DP)** integration via intrinsic Langevin noise (DP-SGLD), rigorously bounding the privacy-utility trade-off using **Rényi Differential Privacy (RDP)**.
- Achieved robust hyperparameter posterior estimation and stable convergence in high-dimensional MuJoCo environments (*Humanoid-v4*, *Ant-v4*) with minimal privacy-utility degradation ($\epsilon \approx 4.0$).
- Theoretically demonstrated **Sub-linear Regret** ($\tilde{O}(\sqrt{dT})$) through Thompson Sampling, significantly outperforming frequentist ϵ -greedy baselines in sample efficiency.
- **Tech Stack:** PyTorch, MuJoCo, SGLD (MCMC), Variational Inference, Differential Privacy (Opacus), TensorBoard.

Stability of Bayesian-SAC under Non-Stationary Reward Scaling

Mar 2026

Implementation of Gaussian Process (GP) surrogates for adaptive entropy control in SAC.

- Mitigated catastrophic policy collapse via **GPyTorch** and **Ray RLLib**; achieved 40% variance reduction.

Academic Activities & Achievements

- **Yonsei AI Club (YAI): Best Academic Member** (Jan 2023); **Student Mentor** (2025–Present) – Awarded **Best Academic Group** (Mar 2026).
- **Samsung AI Challenge 2025: Award Nominee** for Co-Scientist Challenge (A2A Track).

Extracurricular Activities

Republic of Korea Air Force (R.O.K.A.F.) (*Sergeant*)
Squad Leader

Jul 2023 – Jan 2025
Oct 2023 – Nov 2024

- Served as a squad leader for 13 months (twice the standard tenure), managing unit discipline and welfare for 50+ personnel.
- Orchestrated mission readiness and personnel scheduling for a high-responsibility unit for over 13 months.
- Managed complex logistics and crisis response operations, fostering collaborative excellence in a high-stakes environment.

Student Council, Department of Computer Science - Yonsei University

2020–2022

Member, Marketing Team & Task Force Groups — Aug 2020 – Dec 2021

- Designed digital promotional materials (posters, card news) for student events using **Adobe Photoshop** and **Illustrator**.
- Contributed to the **CS Major Guidebook TF**, compiling academic and career guidance for undergraduates
- Participated in the **Student Council Regulation TF**, drafting and revising operational policies for the council.
- Collaborated with multidisciplinary teams to enhance communication between faculty and students.

International Networking & Language Proficiency

- **Technical Networking with OpenAI GTM Team** (Special Invitation)
 - Engaged in technical discussions regarding privacy handling in medical AI services.
 - Demonstrated high-level english proficiency (CEFR C1-C2) and the abilities to discuss emerging AI trends with global industry experts.
- **Languages:** English (Near-native, CEFR C1-C2), Japanese (Controversial), German (Basic, CEFR A1), Spanish/Chinese (Elementary)